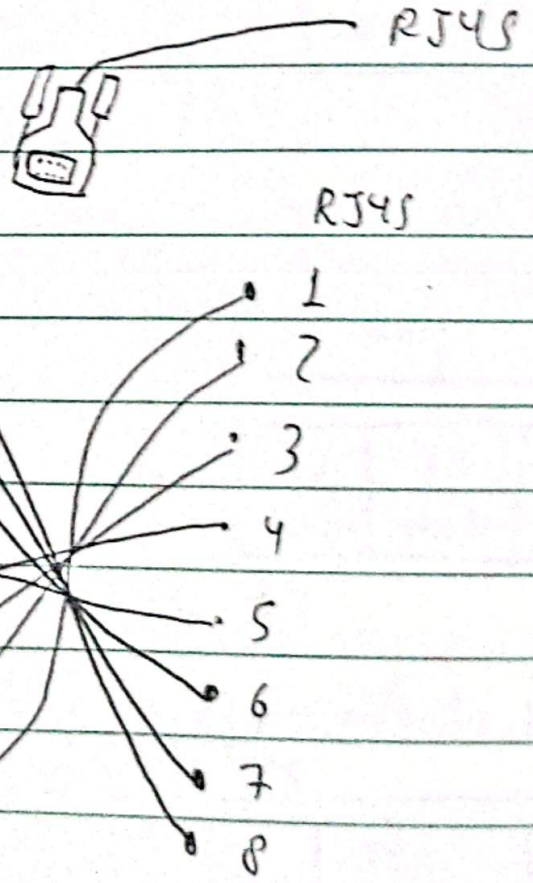
# Introduction to CLI

We can connect device to laptop for configuration through console port like USB mini-B or RJ-45.

Rollover Cable :-




we need to use
↳ Terminal Emulator (PuTTy)

Session ↳ Serial

Data-bits : 8

Stop-bits : 1

Date:..... Time:..... Venue:.....

Parity => None (Used to detect error)

Flow Control (Used to control flow)

Cisco IOS CLI : Router

Router > ]- User ~~Exect~~ EXEC mode

↑ Default Host Name

USER EXEC mode:

(i) Limited

(ii) Can look at something but can't make any changes to configure

(iii) Also called User mode.

Router > enable

Router #

↑ Privileged EXEC mode

Privileged EXEC mode:

(i) Provides complete access to view device's configuration restart the device etc.

(ii) Cannot change the configuration but can change the time on the device. Save conf. file etc



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Date:..... Time:..... Venue:.....

Router>?

↑

Shows Commands

Router # Configuration Terminal

Router (Config)#

This is called **Global Configuration Mode**

# We can protect privilege mode with password.

Router (Config) # enable password CCNA

actual Password

## Configuration files

- There are 2 conf. files kept on device:-
  - running-config / Startup-config at once.

# running-config

↳ The current, active configuration file on the device.

As we enter cmds in the CLI, we edit the active configuration.

Date:..... Time:..... Venue:.....

- Startup-config

The conf. file that will be loaded upon restart of the device.

Syntax:

Router#show running-config

Router# show startup-config

# Saving the Configuration

3 ways to save running-config to make it startup-config

Router#write ]- 1st method.

~~Ou~~ Router. # write memory ]- Second method

Router# copy running-config startup-config

# Service password-encryption

⇒ To strengthen password

Router# conf t

~~Rout~~ Router (config)# service $\underbrace{\text{password-encryption}}$

This will encrypt the password ie CCNA.

Date:..... Time:..... Venue:.....

Example:

Router# show running-config

enable password 7 08026F6028 ] — This pw is also not secure.

Type of Encryption Used

[7 = CISCO Properiatory Encryption Algorithm]

# More Secure Command : $\boxed{\text{enable secret}}$

Router(config)# enable secret Cisco

It's is pw.

Example:-

Router(config)# ~~sh~~ do sh run

enable secret 5 $1$mER..... ] - It's more secure.

enable password 7 08026F6028

∴ Note : If Both pw are same then type 7 encryption will be ignored and enable secret will be valid

Date:..... Time:..... Venue:.....

# Cancelling Commands

Router(config)# no service password-encryption

future password will no longer be encrypted.

Review:

modes
- User Exec mode
- Privileged ~~mode~~ Exec mode
- global conf. mode

Router#> enable (Used to enter privileged mode)

Router# Conf t (used to enter global conf. mode)

~~Router#~~ Router(config)# en password password

(Configures a passwd to protect priviledge exec mode)

Router(config)# service password-encryption

(Encrypts the enable password and other passwd)

Router(config)# enable secret password

(Configures a more secure, always encrypted enable passwd)

Router(config)# run priviledged-exec-level-command

(executes a privileged-exec level cmd from global conf mode)



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Date:..... Time:..... Venue:.....

Router(config)# no command
(removes the command)

Router (config)# show running-config
Router (config)# show startup-config

Router (config)# write
Router (config)# write memory
Router (config)# copy running-config startup-config

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